

|          |          |                                                                                                                                                                        |
|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |          | value.                                                                                                                                                                 |
| <b>3</b> | <b>B</b> | $\Delta p = \text{impulse}$<br>$p_f - p_i = \text{area under F-t graph}$<br>$m(v_f - v_i) = 10(0.5) + 5(0.5)$<br>$3(v_f - 0.50) = 7.5$<br>$v_f = 3.0 \text{ m s}^{-1}$ |
| <b>4</b> | <b>B</b> | Since no net external force acts on the system of 2 particles, final momentum = initial                                                                                |